PHASE 2 – REQUIREMENT ANALYSIS

Requirement Analysis is an important phase in the development of the proposed IT Procurement System. In this phase, the requirements of employees, the IT team, and administrators are identified and documented. The main purpose of the system is to simplify the standard laptop procurement process by reducing manual activities, improving request tracking, and providing a centralized platform for managing laptop requests.

1. Functional Requirements

Functional requirements describe the operations and services that the proposed IT Procurement System must provide. The system should support the complete standard laptop request process, beginning with employee submission and continuing through verification, approval, ordering, and completion. Each function should be clearly defined so that the system can provide a structured and efficient procurement workflow.

Employee Laptop Request Submission

The system should allow employees to submit laptop requests through a simple and user-friendly form. The employee should provide essential information such as employee name, employee ID, department, designation, contact details, reason for requesting a laptop, and required laptop specifications. The system should validate the entered information before accepting the request.

Request Storage and Unique Request Number

After submission, the system should store the request in a centralized database. A unique request number should be generated for every request so that the employee and IT team can identify and track the request easily. Important information such as employee details, request date, laptop requirements, status, approval information, and completion details should be maintained.

Requirement Checking

The system should check the laptop requirements provided by the employee against predefined standard laptop requirements. This helps the organization identify requests that match the approved standard configuration and separate them from requests that may require additional review.

IT Team Request Management

The IT team should be able to view all submitted laptop requests. Authorized IT staff should verify employee information, review laptop requirements, check whether the request meets the standard configuration, and process the request according to organizational procedures.

Request Status Management

The system should provide clear request statuses so that the progress of each request can be monitored. Possible statuses may include New, In Progress, Pending Approval, Approved, Rejected, Ordered, and Completed. Updating the status helps both employees and IT staff understand the current stage of the procurement process.

Request Tracking

Employees should be able to track the progress of their submitted requests using the request number or through their user account. The system should display the current status and relevant updates so that employees do not need to repeatedly contact the IT team for basic status information.

Administrator Management

The administrator should be able to manage the overall procurement process. Administrative functions may include managing users and roles, monitoring requests, maintaining standard laptop requirements, controlling access, reviewing records, and monitoring the overall system workflow.

Search and Filtering

The system should provide search and filtering facilities to help authorized users locate requests quickly. Requests may be searched or filtered using request number, employee name, department, request status, or date.

Request History and Record Maintenance

The system should maintain a history of submitted and processed requests. Important updates and processing activities should be recorded so that the organization has an accurate and centralized record of laptop procurement activities.

2. Non-Functional Requirements

Non-functional requirements describe the quality, performance, security, usability, and reliability expected from the system. These requirements ensure that the IT Procurement System is not only functional but also convenient, secure, dependable, and maintainable.

Easy-to-Use Interface

The system should provide a simple and understandable interface. Forms should contain clearly labelled fields, appropriate instructions, validation messages, and easy navigation so that employees and IT staff can use the system without unnecessary difficulty.

Fast Processing

The system should process requests efficiently and display information within a reasonable response time. Common activities such as submitting a request, viewing request details, searching records, and updating statuses should be completed without unnecessary delay.

Data Security

The system should protect procurement and employee information from unauthorized access. Authentication, authorization, role-based access, and appropriate security controls should be applied to protect stored data.

Reliability

The system should operate consistently and preserve request information accurately. Submitted records should not be lost during normal operations, and important information should remain

available to authorized users.

Scalability

The system should be capable of supporting an increasing number of employees, laptop requests, and records as organizational requirements grow.

Proper Access Control

Different users should have access according to their responsibilities. Employees should manage their own requests, IT staff should process assigned procurement activities, and administrators should have appropriate management privileges.

Accurate Information

The system should maintain correct and consistent information throughout the procurement process. Validation and controlled updates should reduce incorrect or incomplete records.

Easy Maintenance

The system should be designed so that authorized administrators can maintain configurations, user access, standard laptop requirements, workflows, and other settings with minimal difficulty.

Availability

The system should be available to authorized users whenever procurement activities need to be performed, subject to the availability of the underlying ServiceNow platform and network services.

3. Software Requirements

The proposed system can be implemented using the ServiceNow platform because it provides the application development, data management, workflow, notification, access control, and reporting capabilities required for an automated IT procurement process. The following software components are required for developing and accessing the system:

ServiceNow Platform

The ServiceNow platform can be used to create the application, tables, forms, workflows, business rules, notifications, user roles, and reports required for laptop procurement.

Web Browser

A modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox can be used to access the ServiceNow application.

Internet Connection

A stable internet connection is required to access the cloud-based ServiceNow environment and perform procurement activities.

ServiceNow Application Platform

The application platform provides facilities for storing request data, creating user interfaces, managing records, configuring access controls, and automating business processes.

Workflow / Flow Designer

Workflow or Flow Designer capabilities can be used to automate activities such as request routing, approvals, status updates, and notifications.

Notifications

Notification features can be used to communicate important request updates to employees, IT staff, and administrators.

Reporting and Dashboard Features

Reports and dashboards can be used to monitor request volumes, statuses, processing activities, and overall procurement information.

4. Hardware Requirements

The proposed system is web-based, so it does not require specialized hardware for normal users. A standard computer or laptop with internet access is sufficient for accessing the application. The hardware should provide enough performance for smooth browser operation and normal ServiceNow activities.

- Computer or laptop for accessing and managing the application.
- Minimum 4 GB RAM for basic browser and application usage; 8 GB RAM is recommended for smoother multitasking.
- A suitable processor capable of running a modern web browser.
- Monitor or display with sufficient resolution for viewing forms, lists, dashboards, and reports.
- Keyboard and mouse or touchpad for entering and managing information.
- Stable internet/network connectivity for accessing the ServiceNow platform.
- Reliable power supply to avoid interruptions during request processing.
- Sufficient local storage for normal operating system and browser requirements.

5. User Requirements

The proposed system is designed for three major categories of users: employees, IT team members, and administrators. Each user group requires different functions based on its responsibilities in the laptop procurement process.

5.1 Employee Requirements

Employees should be able to access the procurement system using their authorized account and submit a laptop request without unnecessary manual procedures. The employee should enter the required personal and department information, provide the reason for the request, and specify the required standard laptop configuration. After submission, the employee should receive a unique request number and should be able to track the request status. The employee should also be able to view relevant updates until the request is completed.

5.2 IT Team Requirements

The IT team should be able to view and process submitted laptop requests. IT staff should verify the employee information and laptop requirements, identify standard requests, review or process approvals according to organizational procedures, add comments when necessary, and update the request status. The IT team should also be able to search and filter requests, monitor pending work, and access only the information permitted by their assigned role.

5.3 Administrator Requirements

The administrator should be responsible for managing the overall system and procurement process. Administrative functions may include managing users and roles, controlling access, maintaining standard laptop requirements, monitoring requests, configuring workflow activities, reviewing records, and generating reports. The administrator should also be able to monitor the overall performance of the procurement process and ensure that the system remains properly configured.

6. Overall Requirement Summary

The requirement analysis establishes the foundation for the proposed IT Procurement System. The system is intended to provide a centralized and automated method for managing standard laptop requests. Employees can submit and track their requests, the IT team can review and process requests, and administrators can manage the overall procurement workflow and system configuration.

By implementing these requirements, the proposed system can reduce manual effort, improve request visibility, maintain accurate procurement records, provide structured request processing, and improve communication between employees and the IT team. The documented functional, non-functional, software, hardware, and user requirements can be used as a foundation for the next stages of system design, development, configuration, testing, and implementation.

